In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Main), 2017

ALL INDIA TEST SERIES

FULL TEST – II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. A box suspended by a rope is pulled to one side by a horizontal force. The tension in the rope
   (A) Is less than before
   (B) Remains unchanged
   (C) Is greater than before
   (D) May be any of the above, depending on the force applied

2. The co-efficient of kinetic friction between two steel surfaces is
   (A) 0.57 N/kg (B) 0.7 N
   (C) 0.63 N/m (D) 0.6 J/N-m

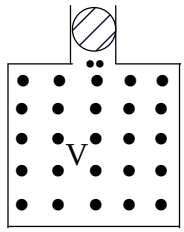
3. The head of an axe has a mass of 2.5 kg. It exerts a force of 80 kN as it penetrates 18 mm into the trunk of a tree. The speed with which the axe head struck the tree was
   (A) 1.2 m/s (B) 34 m/s
   (C) 3.4 m/s (D) 107 m/s

4. A ball strikes a horizontal surface at an angle α with it and rebounds at an angle β, then find the incorrect statement
   (A) β may be equal to α (B) β may be less than α
   (C) β may be greater than α (D) β can not be greater than α

5. The least count of a stop watch is 1/5 second. The time of 20 oscillations of a pendulum is measured to be 25 seconds. The percentage error in the measurement of time will be
   (A) 0.1% (B) 0.8%
   (C) 1.8% (D) 8%

Space for rough work

6. An air chamber of volume V has a neck area of cross – section 'a' into which a ball of mass m can move up and down without friction. If the ball is press down a little and released then find the time period of oscillation of ball. Here air is ideal & its temperature remains constant & the isothermal. Bulk modulus of air is B.



(A) $T= 2\pi\sqrt{2Vm/Ba^2}$ (B) $T= 2\pi\sqrt{Vm/Ba^2}$

(C) $T= 2\pi\sqrt{Vm/2Ba^2}$ (D) None of these

7. A source when at rest in a medium produces waves with a velocity v and a wavelength of λ. If the source is set in motion with a velocity vs what would be the wavelengths produced directly in front of the source?

(A) $\lambda\left(1-\frac{v_s}{v}\right)$ (B) $\lambda\left(1+\frac{v_s}{v}\right)$

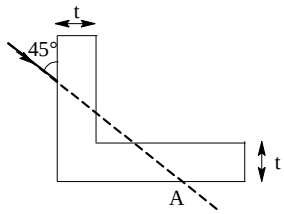
(C) $\lambda\left(1+\frac{v}{v_s}\right)$ (D) $\frac{\lambda v}{v+v_s}$

8. A car just leaves the road when it passes over a hump of radius 40 m. The car's velocity is

(A) 20 m/s (B) 392 m/s

(C) 62 m/s (D) It cannot be found with the mass of the car

9. Two glass slabs of same thickness are joined to form an L as shown. A ray is incident on the combination as shown (in the plane of paper). The final emergent ray will



(A) Come out at point A in the same direction as that of the original ray.

(B) Come out at point A but not in the same direction as that of the original ray.

(C) Does not come out at point A but is in the same direction as that of the original ray.

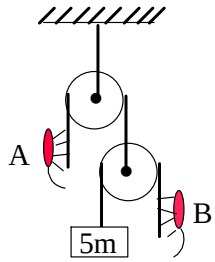
(D) Does not come out at point A and is not in the same direction as that of the original ray.

Space for rough work

10. A solid steel cylinder A and an identical wooden cylinder B are released from two identical inclines from the same height. The incline of A has sufficient friction to ensure rolling and that of B is frictionless, then
(A) A reaches the bottom first
(B) B reaches the bottom with greater Kinetic Energy
(C) Both reach the bottom with same Kinetic Energy
(D) A reaches the bottom with greater Kinetic Energy

11. A 400g piston in a compressor oscillates up and down through a total distance of 80mm. The maximum force on the piston when it moves at 10 cycles/s is
(A) 64 N (B) 68 N
(C) 128 N (D) 132 N

12. Two monkeys A and B of equal mass 'm' can climb strings of a pulley arrangement as shown in figure. Find magnitude of acceleration (in m/s^2) of A with respect to rope so that the block remains stationary. It is given that B is just holding the string. Assume pulley is frictionless and string is massless and inextensible.



(A) 90 m/s^2 (B) 100 m/s^2
(C) 110 m/s^2 (D) 120 m/s^2

13. A boiler in which the gauge pressure is 4 bar develops a leak. The velocity at which the water comes out is
(A) 89 mm/s (B) 28 m/s
(C) 20 m/s (D) 0.8 km/s

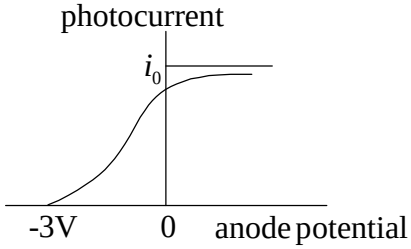
14. A proton (mass *m* and charge *e*) is fired from origin with velocity $\vec{v}=v_0\hat{i}+v_0\hat{j}$ in a uniform magnetic field $\vec{B}=B_0\hat{i}$. In the subsequent motion of the particle the axis of the helical path is

(A) $z=+\frac{mv_0}{eB_0} in\ xz-plane$ (B) $z=+\frac{mv_0}{eB_0} in\ yz-plane$

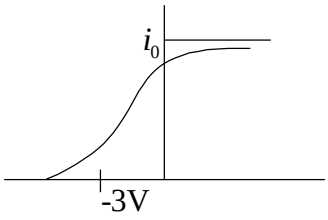
(C) $z=-\frac{mv_0}{eB_0} in\ xz-plane$ (D) $z=-\frac{mv_0}{eB_0} in\ yz-plane$

Space for rough work

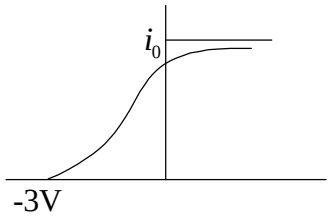
15. Figure shows the anode potential vs photo-current graph when photons of 6eV are incident on cathode in a photoelectric experiment set-up. In the same experimental set-up, if photons of 8eV and same intensity are incident on cathode, then anode-potential *vs* photo-current graph will be (assume 100% efficiency of photons to eject photo-electrons in both cases)



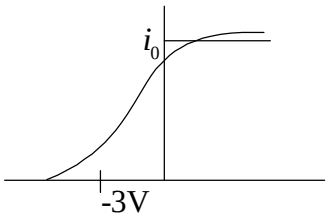
(A)



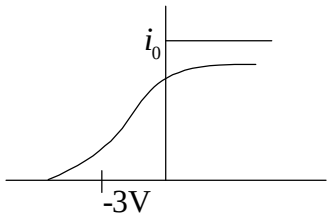
(B)



(C)

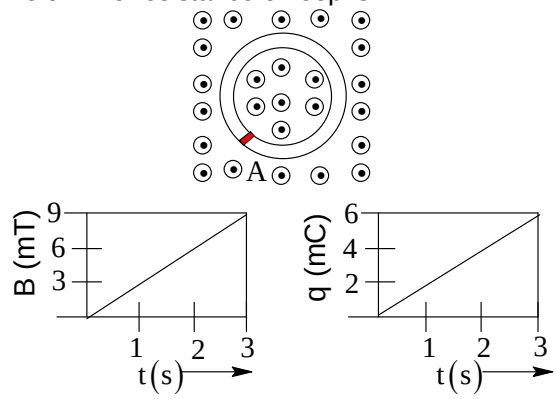


(D)



Space for rough work

16. A uniform magnetic field B increases in magnitude with time 't' as given in fig. 1. A circular conducting loop of area 8 × 10–4 m^2 lies in the field, the field is $\perp$ to the plane of loop. The amount of charge 'q' that passed through the point A on the loop is given in fig. 2 as a function of time 't'. The resistance of loop is:



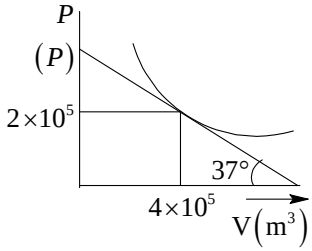
(A) $2.4 m\Omega$ (B) $6 m\Omega$

(C) $1.2 m\Omega$ (D) $4 m\Omega$

17. *P–V* graph for an ideal gas undergoing polytropic process $PV^N =$ constant is shown here. Then N is

(A) 1.5 (B) 2

(C) 1 (D) 3



18. Which of the following combination of units is equal to the watt?

(A) Js (B) $A^2\Omega$

(C) V^2 / A (D) $V^2\Omega$

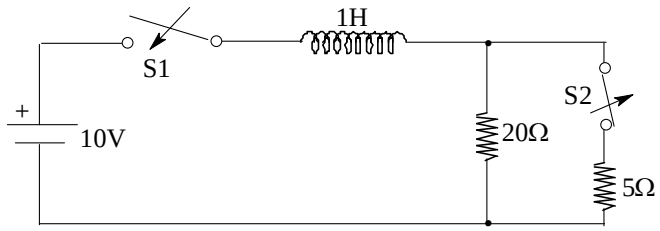
Space for rough work

19. A light bulb rated at 240 V, 150 W is connected to a 120 V power line. The current that flows in it is
(A) 0.313 A (B) 1.25 A
(C) 0.625 A (D) 1.88 A

20. A network of three 5 Ω resistors cannot have an equivalent resistance of
(A) 1.67 Ω (B) 7.5 Ω
(C) 2.5 Ω (D) 15 Ω

21. Frequency of the k_α x-ray for the element calcium (*Ca, z=20) is 8.95*×10^{17}*Hz. Frequency of* k_α x-ray for element cadmium (*Cd*, z = 48) will be
(A) 5.46×10^{18}*Hz* (B) 6.31×10^{18}*Hz*
(C) 5.71×10^{17}*Hz* (D) 6.31×10^{17}*Hz*

22. In the circuit shown, switch S is initially open. At time *t* = 0, S is closed, at this instant the potential difference across the 5 Ω resistor is



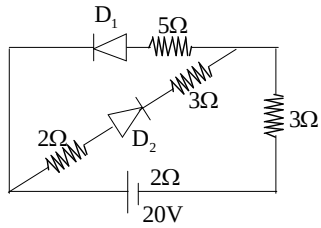
(A) 10 V (B) 2.5 V
(C) 5 V (D) 0 V

Space for rough work

23. An organ pipe of length *L* is open at one end and closed at the other end. The air column in the pipe is vibrating in second overtone. The minimum distance from the open end where the pressure amplitude is half of the maximum value is
(A) $L/4$ (B) $L/8$
(C) $L/12$ (D) $L/15$

24. A long straight wire of circular cross-section is made of a non-magnetic material. The radius of cross-section of the wire is *a*. The wire carries a current *I* which is uniformly distributed over its cross-section. The energy stored per unit length in the magnetic field contained with the wire is
(A) $\mu I^2/2\pi$ (B) $\mu I^2/4\pi$
(C) $\mu I^2/8\pi$ (D) $\mu I^2/16\pi$

25. In the given circuit all diodes are ideal. The current through battery (shown in figure) is
(A) 2 A (B) 1 A
(C) 4 A (D) 4 A



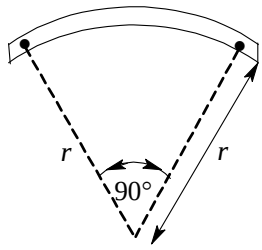
26. At certain time the activity of three radioactive materials are in the ratio of 3: 5:7. What will be the ratio of their activities at any further date?
(A) 1 : 2 : 3 (B) 1 : 1 : 1
(C) 3 : 5 : 7 (D) 7 : 5 : 3

27. Two moles of a monatomic gas is mixed with three moles of a diatomic gas. The molar specific heat of the mixture at constant volume is
(A) 1.9 R (B) 2.1 R
(C) 2 R (D) 1.67 R

Space for rough work

28. A coil of self-inductance L is connected in series with a bulb B and an AC source. Brightness of the bulb increases when:
(A) Number of turns in the coil is increased.
(B) A capacitance of reactance $X_C = X_L$ is included in the same circuit.
(C) An iron rod is inserted in the coil.
(D) Frequency of the AC source is increased.

29. A bar magnet of length '*l*' and magnetic dipole moment 'M' is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be



(A) $\frac{3}{\pi}M$ (B) $\frac{2\sqrt{2}}{\pi}M$

(C) $\frac{M}{2}$ (D) M

30. In a common emitter (CE) amplifier having a voltage gain 12, the transistor used has trans conductance 0.03 mho and current gain 25. If the above transistor is replaced with another one with trans conductance 0.02 mho and current gain 20, the voltage gain will be:
(A) 18 (B) 4
(C) 12 (D) 8

Space for rough work

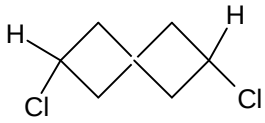
Chemistry

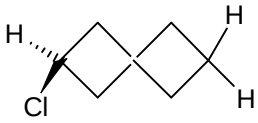
PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. 'A' is 

'B' is 

Which of the following statement is correct about organic compounds A and B ?

(A) Both are optically active

(B) $'A'$ is optically active and B is optically inactive

(C) $'B'$ is optically active and $'A'$ is optically inactive

(D) Both are optically inactive.

2. Dilute solution of KOH reacts with I_2 and forms:

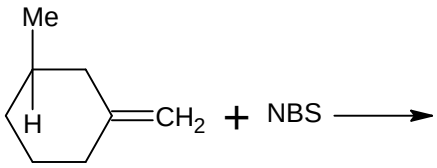
(I) KI (II) KI_3

(III) KIO

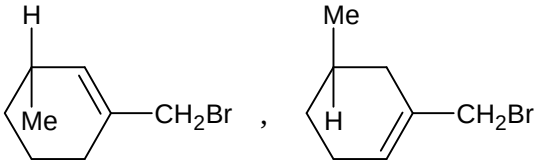
(A) II, II (B) I, III

(C) I (D) I, II, III

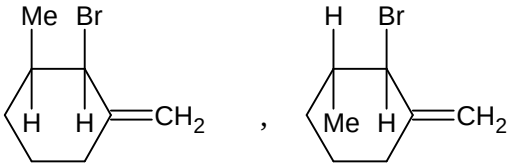
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3. 

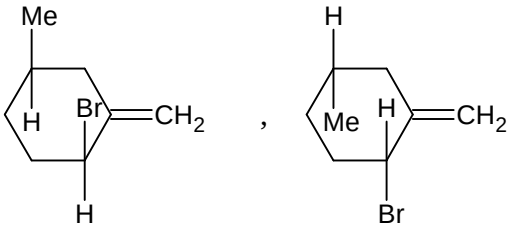
Which of the following statements is incorrect about products?

(A) 

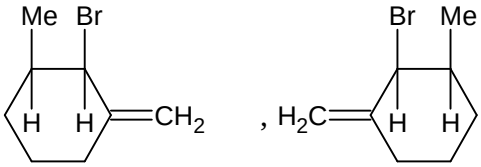
are two possible products.

(B) 

are other two possible products.

(C) 

are other two possible products.

(D) 

are other two possible products.

4. $0.42g$ of Mg_2C_3 reacts with water produces gas X. The gas X combination with O_2 produce CO_2. The volume of CO_2 at STP:

(A) $112mL$ (B) $448mL$

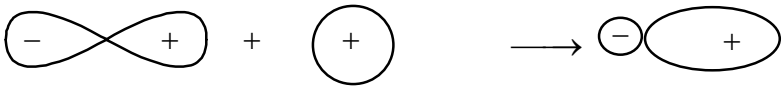
(C) $336mL$ (D) $448mL$

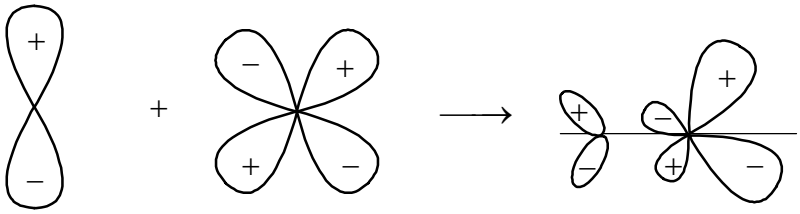
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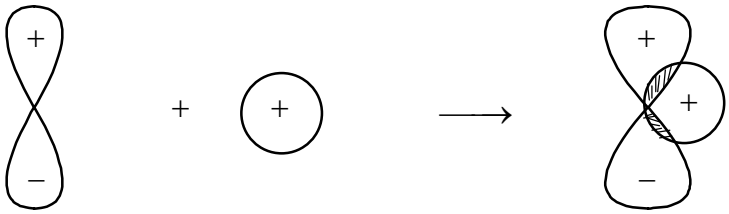
5. Down the group 13 and group14, lower oxidation state compounds are more stable because:
(A) Down the group $I.E.$ decreases
(B) Poor shielding of electrons in $s \& p$ – orbitals as compare to d & f-orbitals
(C) Down the group electronegativity decreases
(D) Outer shell electron are less shielded from nucleus due to presence of electrons in d-orbitals of penultimate shell

6. Which of the following combination of atomic orbitals result into non-bonding?

(A) 

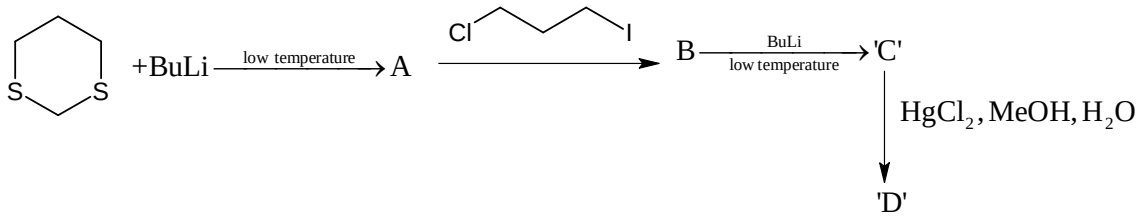
(B) 

(C) 

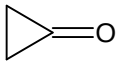
(D) 

Space for rough work

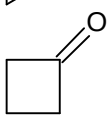
7.



Correct structure of compound 'D' is:

(A) 

(B) CH_3CH_2CHO

(C) 

(D) $CH_3CH_2CH_2CHO$

8. Consider the Vander Waal's equation of state $\left(p+\frac{a}{V^2}\right)(V-b)=RT$ correct statements is:

(A) Value of 'a' of N_2 is larger than that of NH_3 due to higher molecular weight of N_2

(B) Gas 'A' has Vander Wall's constants 'a' $\left(L^2\,atm/mol^2\right)$ and 'b'$\left(L/mol\right)$ as 7 and 4.98×10^{-2}, this gas cannot be liquid at temperature $200°C$ however high pressure may be applied.

(C) At high pressure above equation reduces to $Z=1-\frac{b}{RTV}$

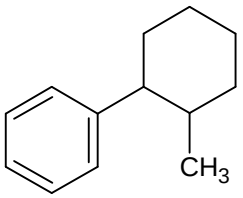
(D) If $Z<1,$ attractive forces are dominating among the gaseous molecules

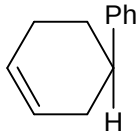
9. cis-2-phenylcyclohexanol $\xrightarrow{Na}$ 'A' $\xrightarrow{CS_2}$ 'B' $\xrightarrow{CH_3I}$ 'C'

'C' is S-methyl xanthate, which on pyrolysis gives:

(A) 3-phenylcyclohexene

(B) 1-phenylcyclohexene

(C) 

(D) 

Space for rough work

10. Bond dissociation enthalpy of gaseous H_2, Cl_2 and HCl are $104, 58$ and 103 $Kcal/mol$ respectively. Then enthalpy of formation of HCl gas is:

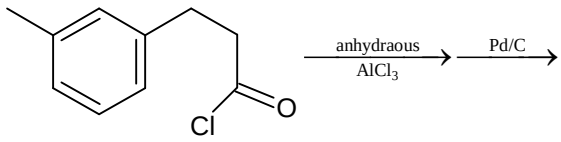
(A) $-22\,kcal$ (B) $+22\,kcal$

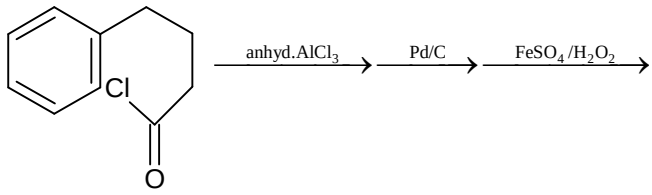
(C) $-44\,kcal$ (D) $+44\,kcal$

11. Which of the following process would result into formation of 7-methyl-1-naphthol?

(A) cis-4-(p-Tolyl)-3-butenoic acid+HF

(B) trans-4-(p-Tolyl)-3-butenoic acid+HF

(C) 

(D) 

12. When a hydrocarbon C_xH_y is burnt, gives $100ml$ of gaseous mixture of CO_2 and H_2O (vapour). On passing product mixture through anhydrous $CuSO_4$ result into decrease of volume by $60\,ml$. Which of the following lowest molecular weight hydrocarbon can satisfy the above condition?

(A) CH_4 (B) C_2H_6

(C) C_2H_4 (D) C_4H_{10}

13. While testing BO_3^{3-}, there is green-edges flame on heating the salt with conc. H_2SO_4 and CH_3OH. Green colour is of

(A) $(CH_3)_3B$ (B) $(CH_3O)_3B$

(C) B_2O_3 (D) H_3Bo_3

Space for rough work

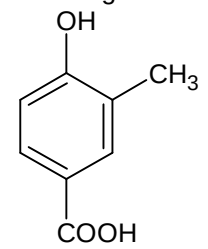
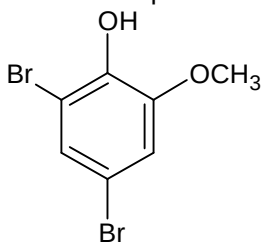
14. Vanillin $(L), C_8H_8O_3$, is isolated from vanilla beans. It give an intense colour with $FeCl_3$ and positive Tollen's test. It is not steam-distilled and does not react with HCl. It goes through following steps:

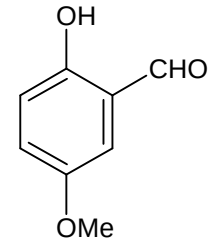
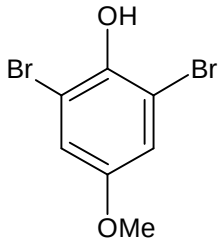
$$L \xrightarrow[1.equivalent\,of\,CH_3I]{OH^-} M,(C_9H_{10}O_3) \xrightarrow{KMnO_4} N,(C_9H_{10}O_4),(HCO_3^- \text{ soluble})$$

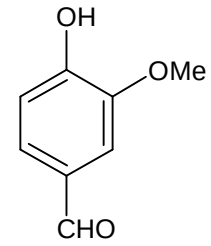
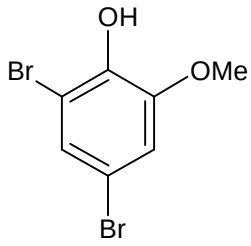
$$N \xrightarrow{conc.\,HI\ \Delta} 3,4\text{-dihydroxy benzoic acid}$$

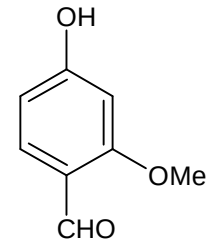
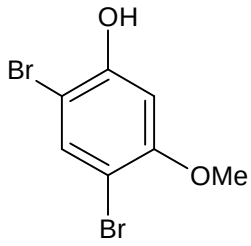
$$L \xrightarrow{[Ag(NH_3)_2]} O \xrightarrow{2\,Br_2} P,(C_7H_6Br_2O_2)$$

Which of the following is correct statement about various products?

(A) L is  and P is 

(B) L is  and P is 

(C) L is  and P is 

(D) L is  and P is 

Space for rough work

15. $[X] + H_2SO_4 \longrightarrow [Y]$, a colourless gas with irritating smell

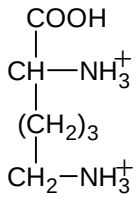
$[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ Green solution

The species [X] and [Y] are

(A) SO_3^{2-}, SO_2 (B) Cl^-, HCl

(C) S^{2-}, H_2S (D) CO_3^{2-}, CO_2

16.



Lysine has p^{ka_1}, p^{ka_2}, p^{ka_3} values 2.18, 8.95 and 10.53 respectively. Then value of its isoelectric point $\left(p^I\right)$ is:

(A) 7.22 (B) 5.565

(C) 6.355 (D) 9.74

17. The colour of potassium dichromate is due to:

(A) d-d transition (B) transition in K^+ ion

(C) ligand-to-metal charge transfer (D) metal to ligand charge transfer

18. The crystal field stabilization energy of high spin d^7 octahedral complex is:

(Where P is pairing energy and Δ_0 is octahedral splitting energy)

(A) $\frac{-4}{5}\Delta_0 + 2P$ (B) $\frac{-4}{5}\Delta_0 + 3P$

(C) $\frac{-9}{5}\Delta_0 + 2P$ (D) $\frac{-9}{5}\Delta_0 + 3P$

Space for rough work

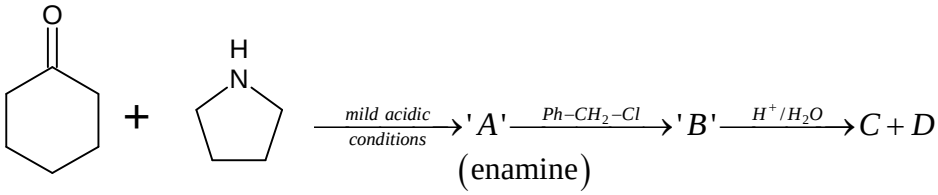
19. Incorrect statement for metal carbonyl $[M(CO)_6]$ is:

(A) σ bonding electrons are donated by CO group to the metal.

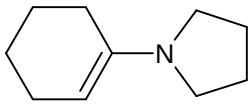
(B) $d\pi - p\pi$ back bonding occurs from the filled metal orbitals.

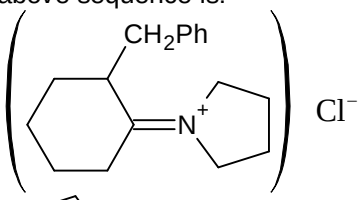
(C) $d\pi - p\pi$ back bonding makes M-CO bond stronger.

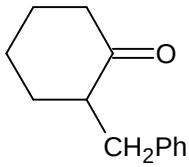
(D) C – O bond length is greater in CO molecules as compare to C – O bond length in given complex.

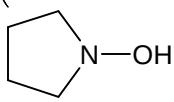
20. 

The incorrect statement about various organic products in above sequence is:

(A) A is 

(B) B is 

(C) C is 

(D) D is 

21. A 0.001 molal solution of a complex represented as $Pt(NH_3)_4Cl_4$ in water had freezing point depression of 0.0054^0C. Given K_f for H_2O = 1.86 K molality^{-1}. Assuming 100% ionisation of the complex. Which of the following is incorrect statement about the complex?

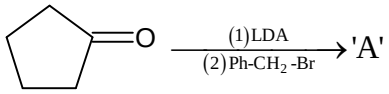
(A) Formula of Complex is $[Pt(NH_3)_4]Cl_4$

(B) 2 Cl atoms are attached as ligands in the complex

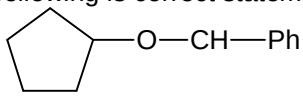
(C) The value of Van't Hoff factor is 3

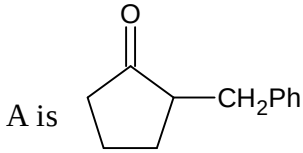
(D) The complex is octahedral in shape

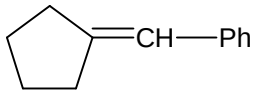
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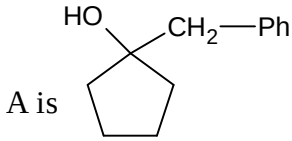
22. 

Which of the following is correct statement about compound 'A' :

(A) A is 

(B) A is 

(C) A is 

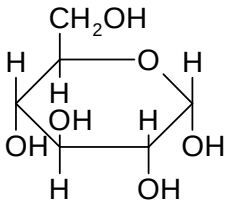
(D) A is 

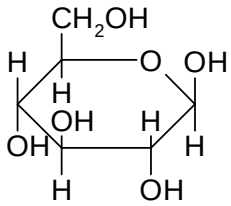
23. An orange coloured compound (A) when treated with $NaCl$ & conc. H_2SO_4 gives red brown vapour (B). This vapour when passed through $NaOH$ gives yellow coloured solution (C). On addition of CH_3COOH and $(CH_3COO)_2 Pb$ to this yellow solution, yellow colured ppt (D) is obtained. Which of the following compound is not resemble with above information:

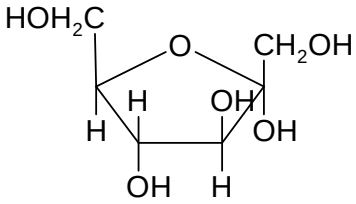
(A) 'A' is $K_2Cr_2O_7$ (B) 'B' is CrO_2Cl_2

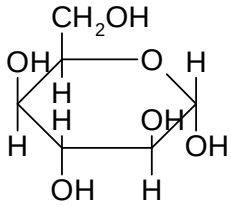
(C) 'C' is Na_2CrO_4 (D) 'D' is $PbCl_2$

24. Which of the following is correct Haworth projection for α-D glucopyranose?

(A) 

(B) 

(C) 

(D) 

Space for rough work

25. Adsorption of gases on solid surface is generally exothermic because:
(A) chemical bonds are always formed
(B) adsorbate decreases the surface energy of adsorbent
(C) entropy increases
(D) free energy increases

26. Which of the following statement is false, if 'a' is the unit cell edge length?
(A) $0.866a$ is the shortest distance between cation and anion in $CsCl$ crystal.
(B) $0.707a$ is the shortest distance between two cations in CaF_2 crystal
(C) $0.433a$ is the shortest distance between carbon atom in diamond
(D) $0.866a$ is the shortest distance between two cation in rock salt crystal

27. $2A+C \longrightarrow P$ follows the mechansm as shown below

$2A \rightleftharpoons B \quad K\ (\text{equilibrium constant})$

$B+C \longrightarrow P \quad K_f\ (\text{rate constant})$

The overall rate, $+\frac{d[P]}{dt}$ based on above mechanism is given by

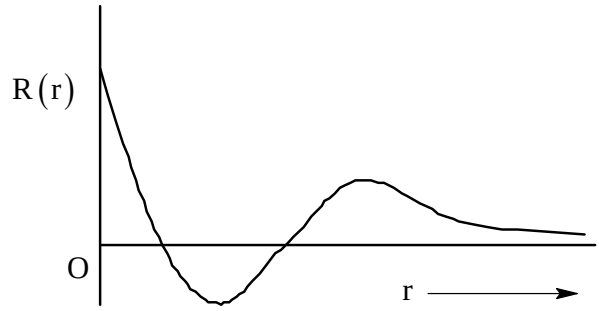
(A) $KK_f[A]^2[C][B]$ (B) $KK_f[A]^2[C]$
(C) $K[A][B]$ (D) $K_f[B][C]$

28. A current of $0.55\,\text{Amp}$ deposits $0.55\,\text{gm}$ of a metal in 100 minutes. Calculate atomic mass of the metal if equivalent weight $= \frac{M.W.}{3}$?

(A) 100 (B) 45
(C) 48.25 (D) 144.75

Space for rough work

29.



The radial part of wave function of an orbital is plotted against distance from nucleus. Which orbital represent above graph?

(A) $1s$ (B) $2s$

(C) $3s$ (D) $2p$

30. pH of $0.01M-(NH_4)_2SO_4$ and $0.02M-NH_4OH$ buffer $\left(p^{Ka}\ of\ NH_4^+ = 9.26\right)$ is:

(A) 9.26 (B) $9.26+\log 2$

(C) $9.26-\log 2$ (D) $4.74+\log 2$

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. If $n(s)$ represents the number of elements in the set S, such that $n(A)=10$ and $n(B)=20$, then the value of $n(A\Delta B)$ cannot be:

   (A) 30 (B) 10
   (C) 8 (D) 12

2. If $\sqrt{(x^2+2x+1)}+2^{f(x)}=4$, then the maximum value of $f(x)$ is:

   (A) 1 (B) 2
   (C) 3 (D) 4

3. The number of solution(s) of $\max\left\{2^{-x},2^{x},\frac{1}{2}\right\}=\frac{3}{4}-|x|$ is:

   (A) 1 (B) 0
   (C) 2 (D) Infinitely many

4. Let $\frac{1}{3}+\frac{2}{2.3.4}+\frac{2}{3.4.5}+....+\frac{2}{n(n+1)(n+2)}=\frac{n(n+3)}{k(n+1)(n+2)}\forall nfN$, then the value of k is:

   (A) 1 (B) 2
   (C) 3 (D) 4

Space for rough work

5. A student preparing for IIT-JEE at FIITJEE is given 6 targets numbered to 1 to 6 of each subject physics, chemistry and mathematics. He or she keeps all his books in different shelves (for each subject) in order, number of ways to select all 18 books one by one such that books of each subject must be selected in order (i.e. all books of one subject to the selected one after another, then of second subject and then of third subject) is:

(A) $\frac{18!}{6!6!6!}$ (B) $\frac{18!}{6!6!6!}\times 3!$

(C) $\frac{6!6!6!}{3!}$ (D) $\frac{18!}{6!6!6!3!}$

6. If x is such that $\left[2\sin x\right]+\left[\cos x\right]=-3$, for some $x\in\left[0,25\right]$ then range of the function $g\left(x\right)=\sin x+\cos x,$ where $\left[.\right]$ represents greatest integer function is:

(A) $\left[-\sqrt{2},-1\right]$ (B) $\left[-\sqrt{2},-1\right)$

(C) $\left(-\sqrt{2},\frac{-1-\sqrt{3}}{2}\right)$ (D) $\left[\frac{-1-\sqrt{3}}{2},-1\right]$

7. If $f\left(x\right)=\left\{x\right\}+\left\{-x\right\}$ and $g\left(x\right)=\sin^{-1}x$, then number of elements in the range of $gof\left(x\right)$ is (where $\left\{.\right\}$ represents fractional part function):

(A) 0 (B) 1

(C) 2 (D) 3

8. The number of real solution of $\left(x,y\right)$ where $\left|y\right|=\cos x$ and $y=\cos^{-1}\left(\cos x\right),-2\pi\le x\le 2\pi$ is/are:

(A) 1 (B) 2

(C) 3 (D) 4

Space for rough work

9. Let $f(x)=[x+2]$ and $g(x)=\cos x$, where $[.]$ denotes the greatest integer function. The value of $(fog)'\left(\frac{\pi}{2}\right)$, is:

(A) 0 (B) 1
(C) -1 (D) Does not exist

10. Number of points of non-differentiability of $f(x)=\min\{0,\cos x,\sin x\}$ in $(0,2n\pi)$ is:

(A) $3n-1$ (B) $3n$
(C) $3n+1$ (D) $3n+2$

11. If $y=\cos x$ then $\frac{d^2}{dy^2}(\sin^5 x)$ is:

(A) $15\sin x+20\sin^3 x$ (B) $15\sin x-20\sin^3 x$
(C) $20\sin x+15\sin^3 x$ (D) $20\sin x-15\sin^3 x$

12. Find the maximum possible area of rectangle such that it two vertices are on y-axis and other two vertices on the lines $y=x$ and $3x+2y=6$ respectively and the rectangle is inscribed in triangle, $y\geq 0,\ y\geq x,\ 3x+2y\leq 6$ is:

(A) $\frac{3}{10}$ (B) $\frac{7}{10}$
(C) $\frac{9}{10}$ (D) $\frac{11}{10}$

13. If the line $\ell x+my+n=0$ is a normal to the curve $xy=1$, then possible value of ℓ and m is:

(A) $\ell=2,m=1$ (B) $\ell=2,m=-1$
(C) $\ell=-2,m=-1$ (D) $\ell=-1,m=-2$

Space for rough work

14. In a ΔPQR (with usual solutions) if $\begin{vmatrix} 1 & p & q \\ 1 & q & r \\ 1 & r & p \end{vmatrix} = 0,$ then the value of $\cos 2P + \cos 2Q + \cos 2R$ is:

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

15. The number of values of $'t'$ for which the system of equations $(t+1)x + 8y = 4t,\ tx + (t+3)y = 3t - 1$ has no solution is:

(A) 0 (B) 1

(C) 2 (D) 3

16. If $4x - ay + 3z = 0,\ x + 2y + az = 0$ and $ax + 2z = 0$ have a non-trivial solution, then the value of a is:

(A) 0 (B) 1

(C) 2 (D) -0

17. $\int \frac{3x^2 + 2x}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} dx$ is equal to:

(A) $\frac{1}{2}\tan^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$ (B) $\frac{1}{2}\sin^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$

(C) $\tan^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$ (D) $\sin^{-1}\left(\frac{x^3 + x^2 + 1}{2}\right) + C$

Space for rough work

18. If $f(x)=\int_{\frac{1}{x}}^{\sqrt{x}}\sin\left(t^2\right)dt,$ then $f'(1)$ is equal to:

(A) $\frac{3}{2}\sin 1$ (B) $\frac{1}{2}\sin 1$

(C) $\sin 1$ (D) $2\sin 1$

19. Area bounded by the curves $y=\left[\frac{x^2}{36}+1\right], y=x-2$ and $x=0$ above x-axis is, where $[.]$ denotes greatest integer function:

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) $\frac{5}{2}$ (D) $\frac{7}{2}$

20. If the equation represented by the solution of the differential equation $(xy-1)\frac{dy}{dx}+y^2=0$ passes through the points $(2,1)$ and (k,e), then the value of k is:

(A) e^2 (B) $\frac{1}{e}$

(C) $\frac{2}{e}$ (D) $\frac{3}{e}$

Space for rough work

21. Total number of unit vectors in $xy-$plane making on angle of $45°$ with the vector $\hat{i}+\hat{j}$ and an angle of $60°$ with the vector $3\hat{i}-\hat{j}$ is:

(A) 0 (B) 1

(C) 2 (D) 4

22. If a and b are selected from the set $\{1,2,3,.....10\}$, then probability that a^2-b^2 is divisible by 3 is:

(A) $\frac{1}{15}$ (B) $\frac{4}{15}$

(C) $\frac{8}{15}$ (D) $\frac{13}{15}$

23. The value of sum of the series ${}^{13}C_0.{}^{14}C_1+{}^{13}C_1.{}^{14}C_2+{}^{13}C_2.{}^{14}C_3+.....+{}^{13}C_{13}.{}^{14}C_{14}$ is:

(A) ${}^{27}C_{12}$ (B) ${}^{27}C_{14}$

(C) ${}^{27}C_{16}$ (D) ${}^{27}C_{10}$

24. The distance of the point $(2,3)$ from the line $2x-3y+9=0$ measured along a line parallel to $x-y+1=0$ is:

(A) $4\sqrt{2}$ (B) $3\sqrt{2}$

(C) $2\sqrt{2}$ (D) $\sqrt{2}$

25. If the two circles $x^2+y^2=1$ and $x^2+y^2-8x-6y+\lambda^2=0$ have exactly two common tangents then the number of integral value of λ is:

(A) 0 (B) 1

(C) 3 (D) 5

Space for rough work

26. If PQ and RS are two perpendicular chord of the rectangular hyperbola $xy=1$. If C is the centre of this hyperbola then product of the slopes of CP, CQ, CR and CS is:

(A) 0 (B) 1

(C) -1 (D) Not defined

27. If z_1, z_2 and z_3 are the vertices of ΔABC, taken is anticlockwise order such that $\angle B = \frac{\pi}{4}$ and $BC = \sqrt{2}(AB)$, then $i(z_2 - z_3)$ is equal to:

(A) $z_1 + z_2 + z_3$ (B) $z_2 + z_3 - z_1$

(C) $z_1 + z_3 - 2z_2$ (D) $z_2 + z_3 - 2z_1$

28. 6 person A, B, C, D, E and F are to be seated at a circular table. In how many ways can this be done if A must always has either B or C on his right and B must always has either C or D on his right:

(A) 8 (B) 12

(C) 18 (D) 30

29. Standard deviation of two distributions are same and their arithmetic means are 30 and 25 respectively. If their coefficient of variation is v_1 and v_2, then $6v_1 - 5v_2$ is equal to:

(A) 0 (B) 1

(C) 2 (D) 3

30. If A, G and H are arithmetic, geometric and harmonic means between three given numbers a, b and c then the equation having a, b and c as its roots is:

(A) $x^3 + 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$ (B) $x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0$

(C) $x^3 - 3Ax^2 + \frac{3G^3}{H}x + G^3 = 0$ (D) $x^3 + 3Ax^2 - \frac{3G^3}{H}x + G^3 = 0$

Space for rough work